

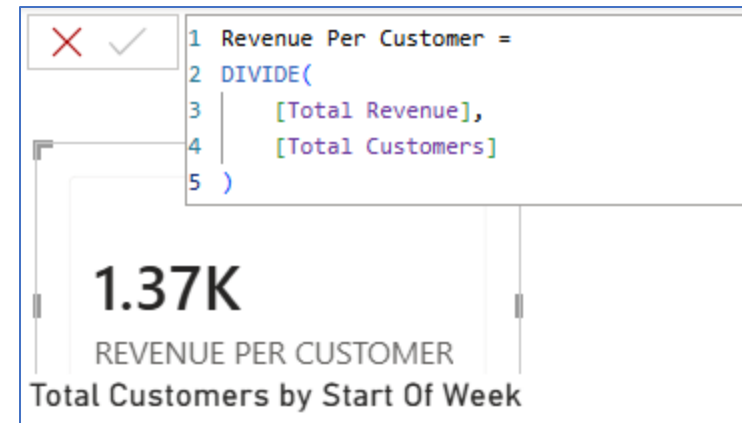
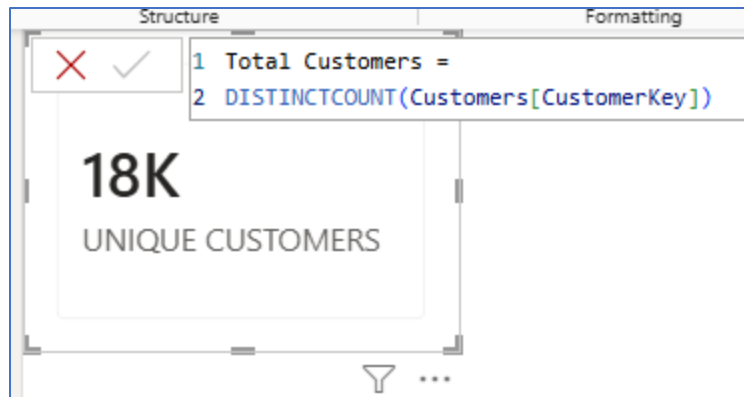
Charts & Visualization

Name: Chauhan Sumit

Dept: Data Engineering

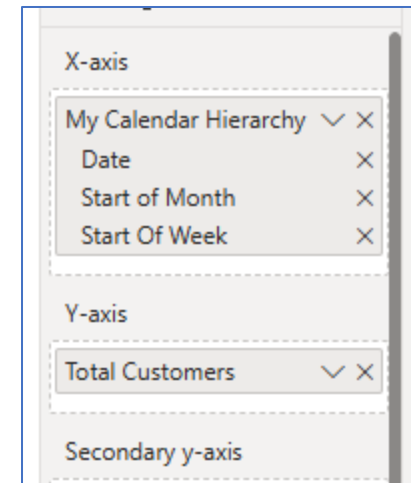
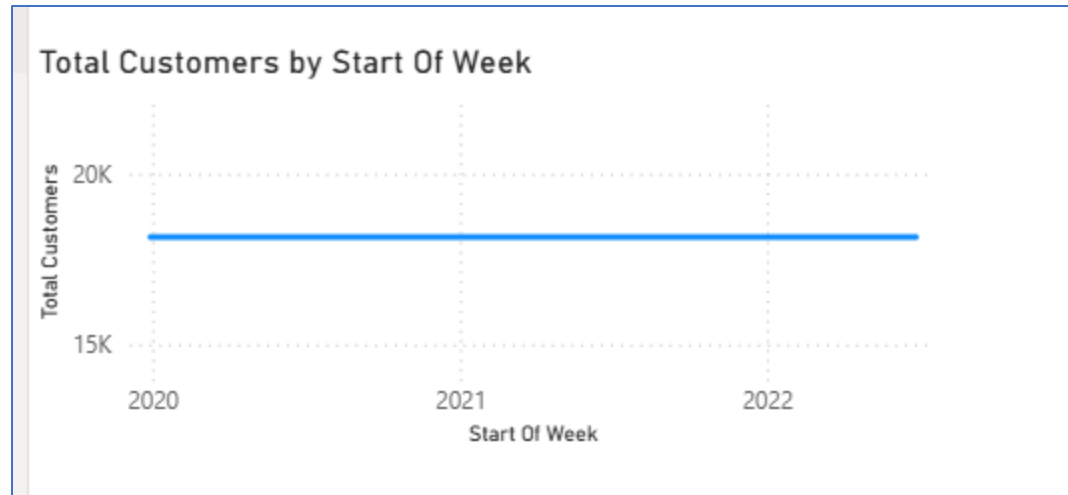
a) Cards

- Insert a card in the Customer Detail report page to show Total Customers, and rename the field “UNIQUE CUSTOMERS”
- Copy and paste to create a second card showing Average Revenue per Customer, and rename the field “REVENUE PER CUSTOMER”



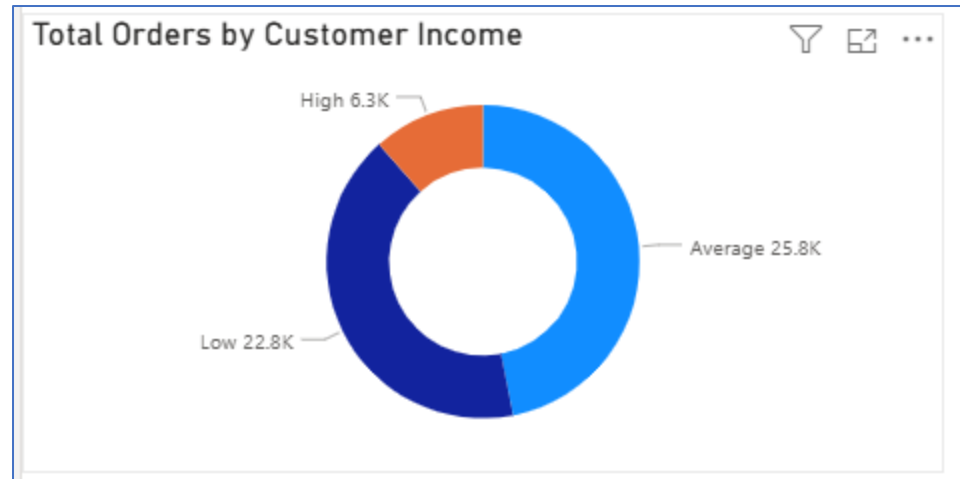
b) Line Charts

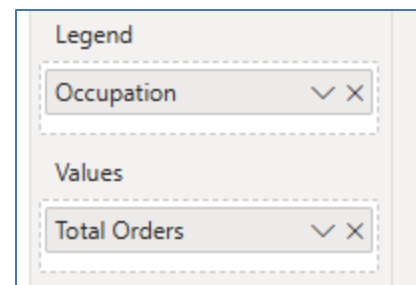
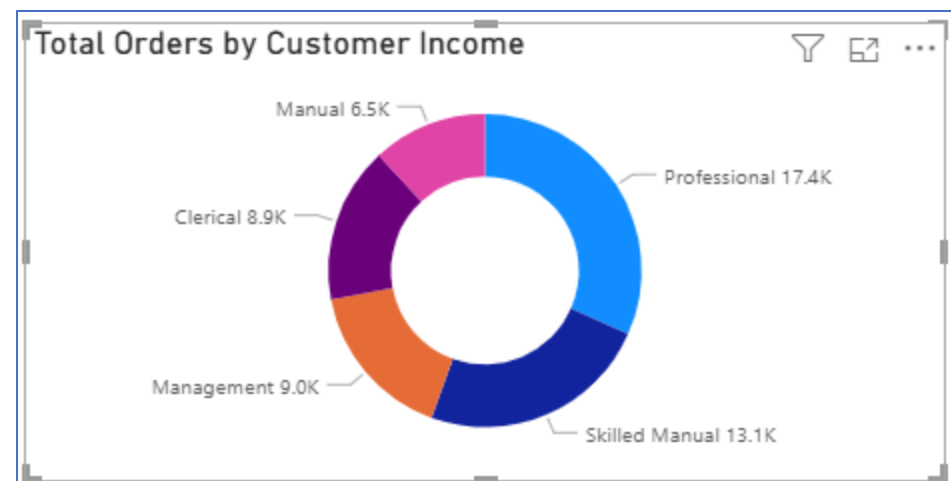
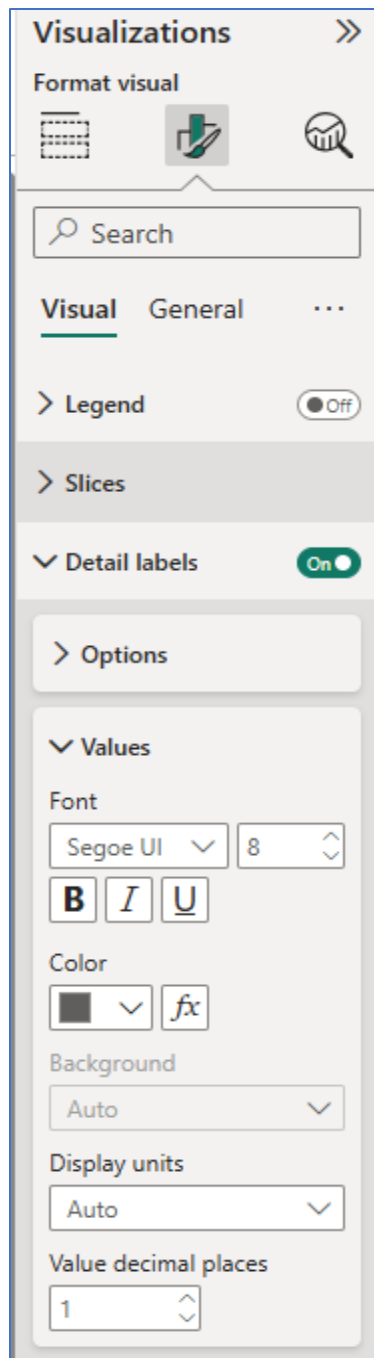
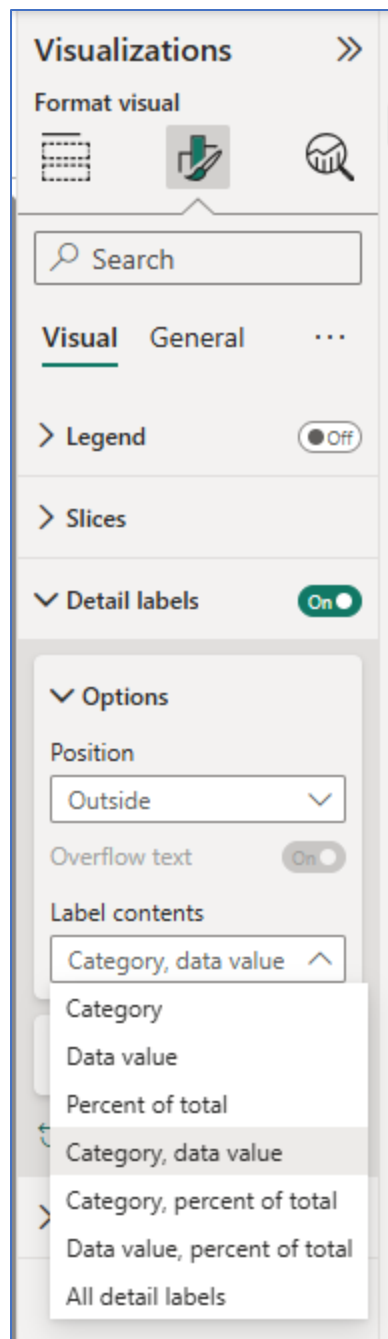
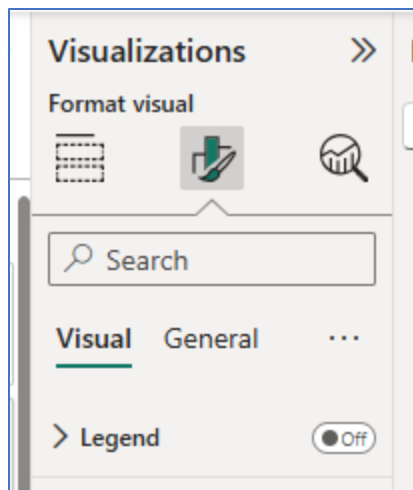
- Add a line chart to the Customer Detail report showing Total Customers by week
- Add a trend line and a zoom slider to the x-axis



c) Donut Chart

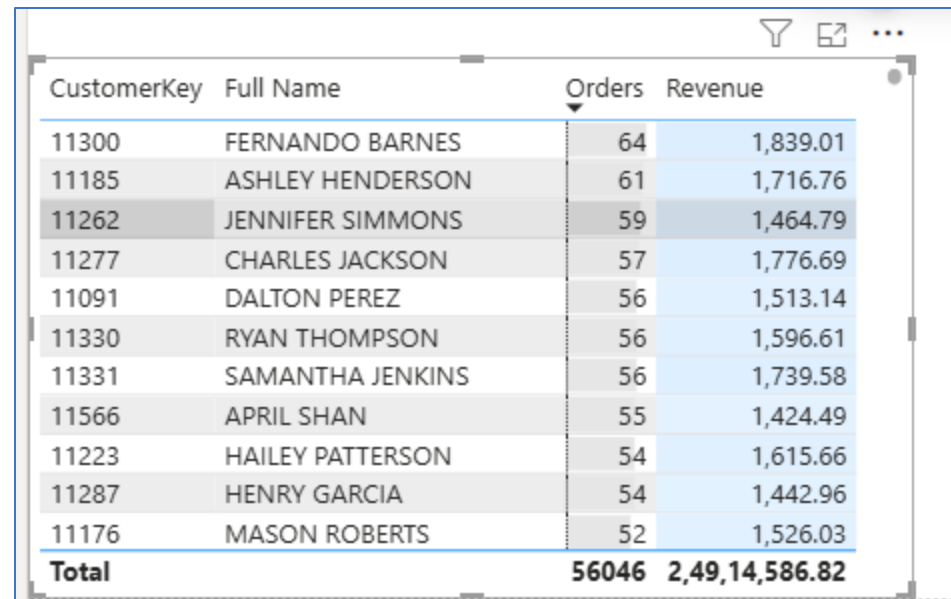
- Add a donut chart to the Customer Detail report showing Total Orders by Income Level
- Add a chart title, turn off the legend, and update the data labels to show the category and value (font size 8, 1 decimal place)
- Add a visual-level filter to exclude customers with a “Very High” income level
- Copy the chart to show Total Orders by Occupation, and add a visual-level filter to display the three occupations with the most orders





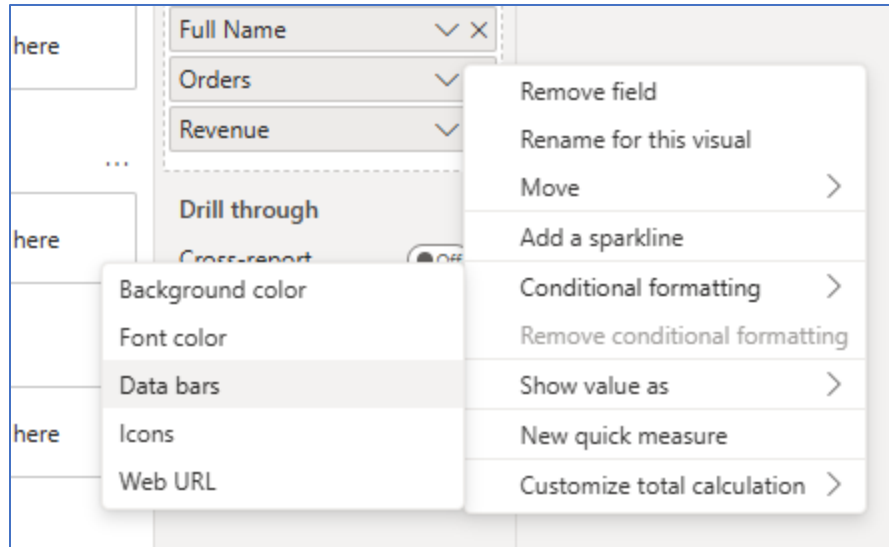
d) Tables

- Add a table to the Customer Detail report to show Customer Key, Full Name, Total Orders (as “Orders”) and Total Revenue (as “Revenue”)
- Use conditional formatting to add light gray data bars to the orders column and a white > blue color scale to the revenue column
- Add a visual-level filter (Top N) to show the 100 customers with the most orders, and add a chart title (“Top 100 Customers”)
- Sort the table descending by orders



The screenshot shows a table visualization in a software interface. The table has four columns: 'CustomerKey', 'Full Name', 'Orders', and 'Revenue'. The 'Orders' column is highlighted with light gray data bars, and the 'Revenue' column is highlighted with a white-to-blue color scale. The table is sorted by 'Orders' in descending order. At the bottom, there is a 'Total' row. The interface includes a filter icon, a refresh icon, and a menu icon in the top right corner.

CustomerKey	Full Name	Orders	Revenue
11300	FERNANDO BARNES	64	1,839.01
11185	ASHLEY HENDERSON	61	1,716.76
11262	JENNIFER SIMMONS	59	1,464.79
11277	CHARLES JACKSON	57	1,776.69
11091	DALTON PEREZ	56	1,513.14
11330	RYAN THOMPSON	56	1,596.61
11331	SAMANTHA JENKINS	56	1,739.58
11566	APRIL SHAN	55	1,424.49
11223	HAILEY PATTERSON	54	1,615.66
11287	HENRY GARCIA	54	1,442.96
11176	MASON ROBERTS	52	1,526.03
Total		56046	2,49,14,586.82



Data bars - Orders

Format style: Data bars

Apply to: Values only

What field should we base this on?: Orders

Bar direction: Left to right

Minimum: Lowest value

Maximum: Highest value

Positive bar: ☐

Negative bar: ☐

Axis: ☐

☐ Show bar only

[Learn more about conditional formatting](#)

OK Cancel

CustomerKey ^ 🔒

top 100 by Total O... 🔗 👁

Filter type ⓘ

Top N ▼

Show items

Top ▼ 100

By value

Total Orders ×

Apply filter

Top 100 Customers			
CustomerKey	Full Name	Orders	Revenue
11300	FERNANDO BARNES	64	1,839.01
11185	ASHLEY HENDERSON	61	1,716.76
11262	JENNIFER SIMMONS	59	1,464.79
11277	CHARLES JACKSON	57	1,776.69
11091	DALTON PEREZ	56	1,513.14
11330	RYAN THOMPSON	56	1,596.61
11331	SAMANTHA JENKINS	56	1,739.58
11566	APRIL SHAN	55	1,424.49
11223	HAILEY PATTERSON	54	1,615.66
11287	HENRY GARCIA	54	1,442.96
Total		2669	4,37,347.46

e) Top Ncards

- Add a card to the Customer Detail report to show Full Name
- Add a visual-level filter (Top N) to show the top customer (Full Name) in terms of Total Revenue
- What do you notice when you filter the report for low income customers?
- How could you modify the Top N filter to correct this?

Full Name	
JORDAN TURNER	

Filters on this visual ...

Full Name

top 1 by Total Reve...

Filter type ⓘ

Top N ▼

Show items

Top ▼ 1

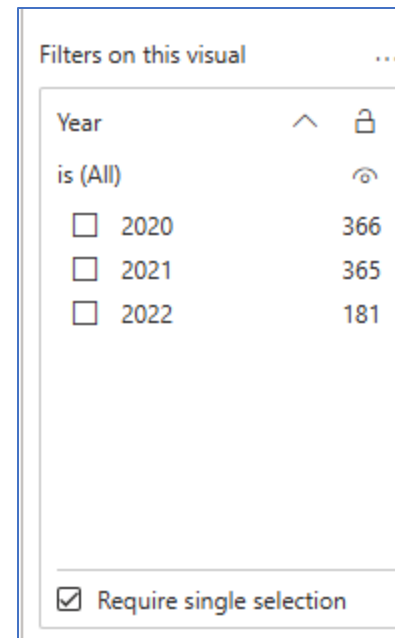
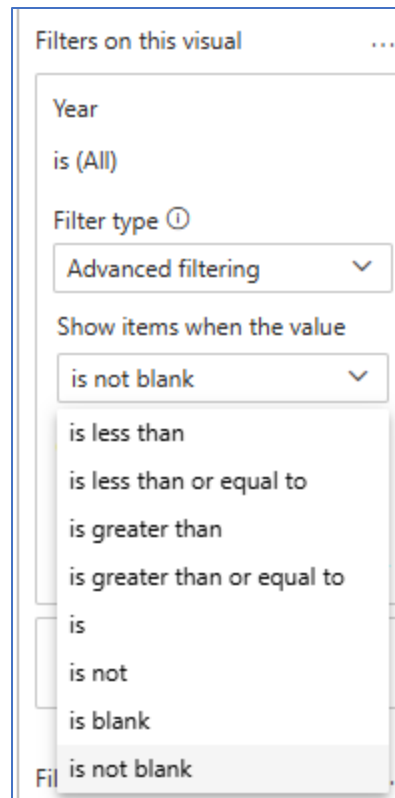
By value

Total Revenue ×

Apply filter

f) Slicers

- Add a slicer to filter the Customer Detail report page by Year
- Add a visual-level filter to exclude blanks
- Choose any slicer style that allows users to filter individual years or across multiple years



352.93

REVENUE PER CUSTOMER

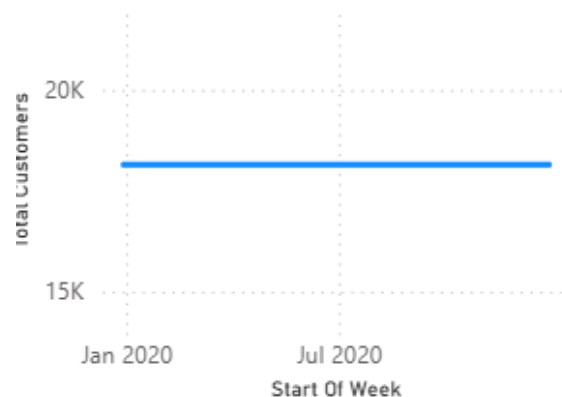
18K

UNIQUE CUSTOMERS

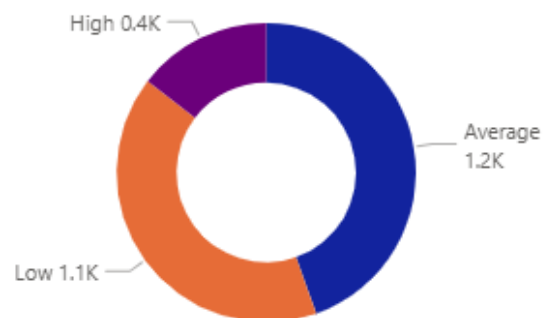
Top Customer By revenue

RAFAEL XU

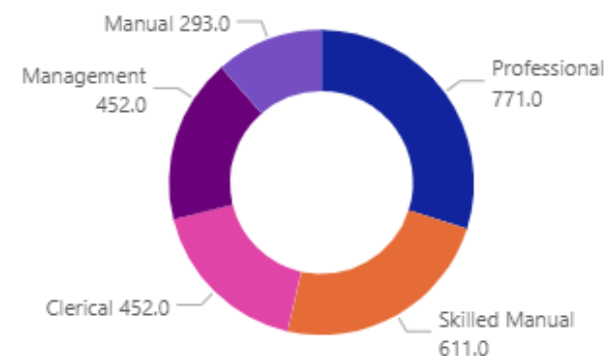
Total Customers by Start Of Week



Total Orders by Customer Income

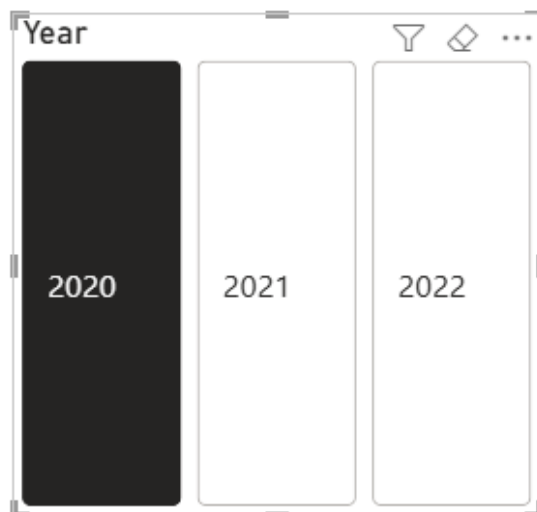


Total Orders by Customer Income



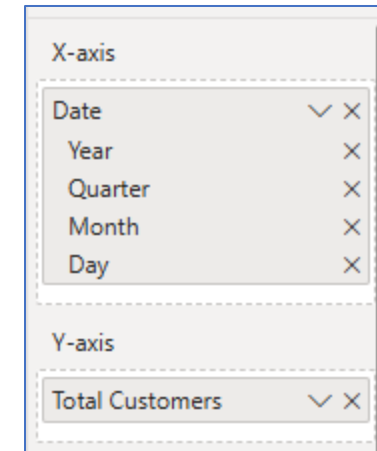
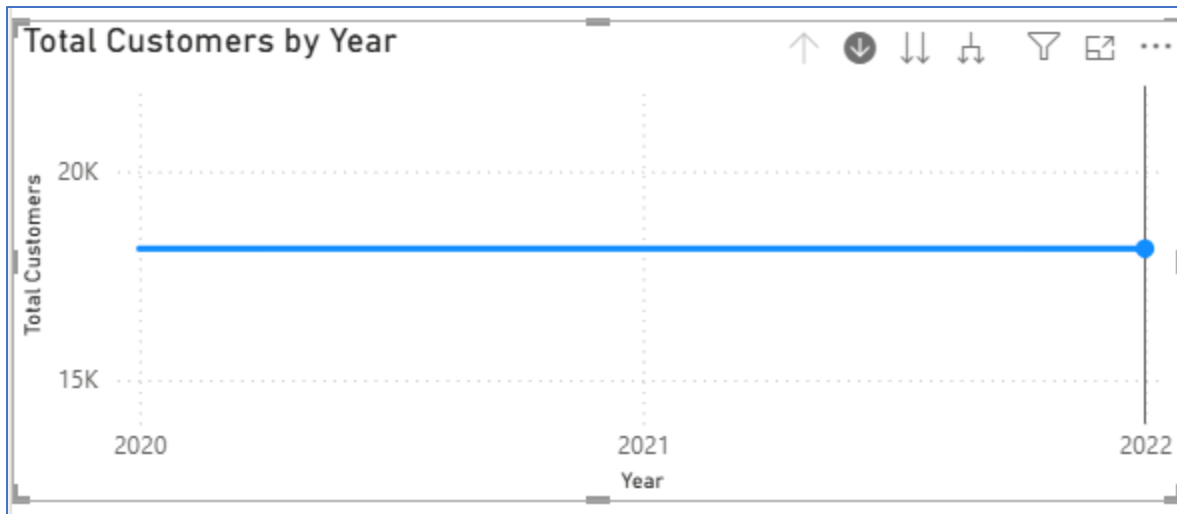
Top 100 Customers

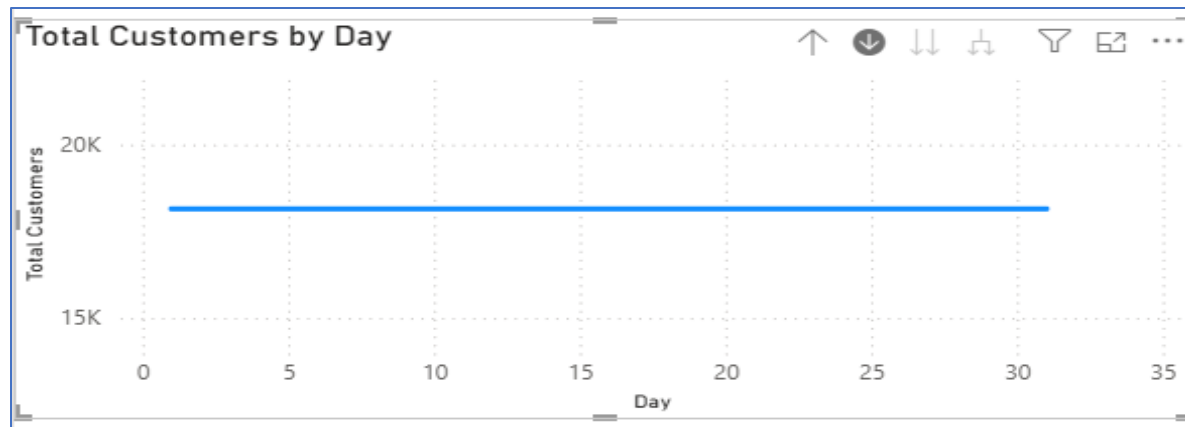
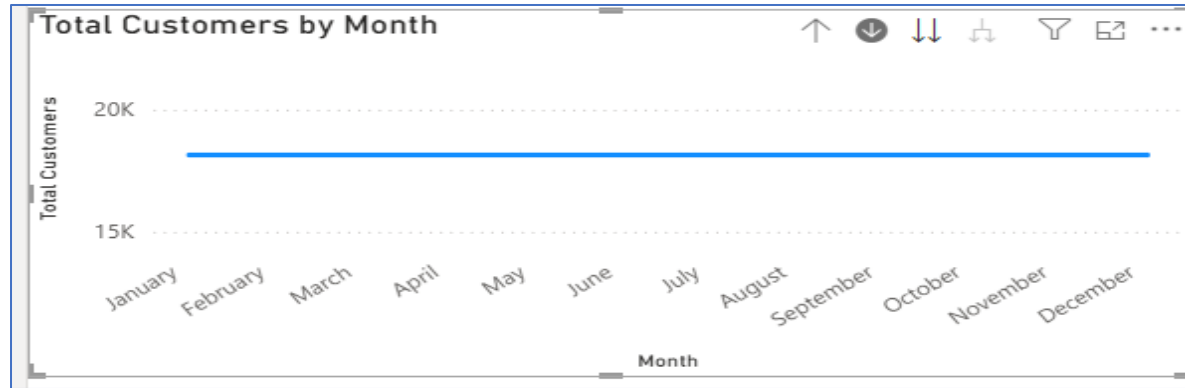
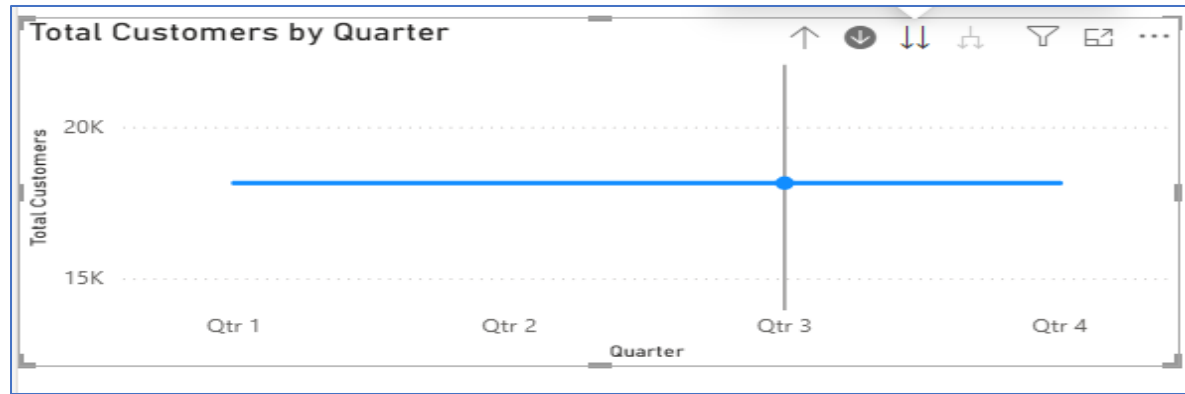
CustomerKey	Full Name	Orders	Revenue
11090	TREVOR BRYANT	1	3,578.27
11129	JULIA WRIGHT	1	3,578.27
11171	JONATHAN HILL	1	3,578.27
11175	LUIS WANG	1	3,578.27
11191	KRISTI PEREZ	1	3,578.27
11216	JASMINE TORRES	1	3,578.27
11217	NATALIE ADAMS	1	3,578.27
11224	TIFFANY LI	1	3,578.27
11227	MARSHALL CHAVEZ	1	3,578.27
11237	CLARENCE ANAND	1	3,374.90
Total		2630	64,04,933.58



g) Drill Down

- In the Customer Detail report, update the X-axis of the line chart to pull in Date Hierarchy
- Use the chart header to drill up and drill down to explore trends at each level of granularity
- Test drill mode to change the granularity by selecting individual data points in the chart
- Turn off drill mode and show the chart at a weekly level of granularity by default





h) Report Interactions

- On the Customer Detail tab, edit the report interactions based on the following logic:
 - When a filter is applied to the line chart, the donut charts should filter (not highlight)
 - When a filter is applied to the table, the line chart and donuts should not filter
 - The slicer should filter all visuals on the report page

352.93

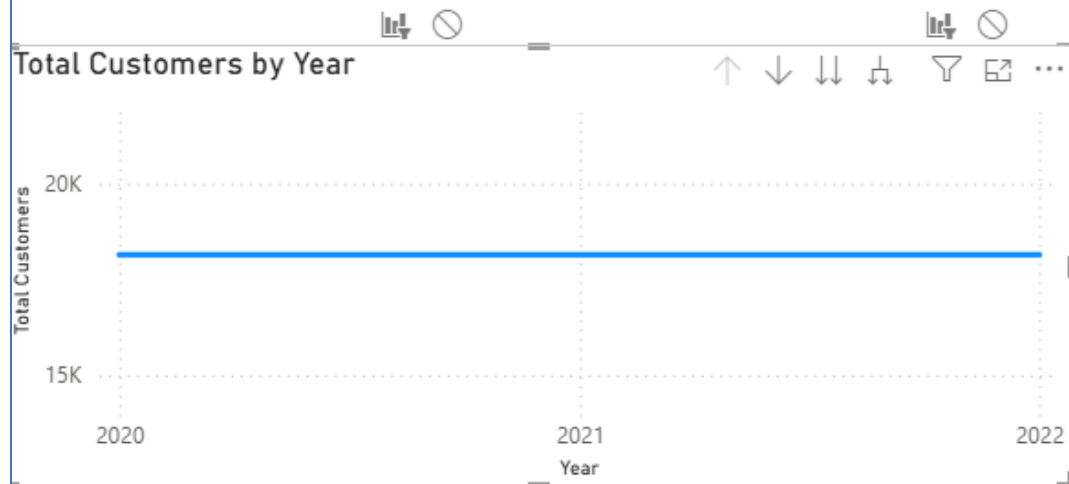
REVENUE PER CUSTOMER

18K

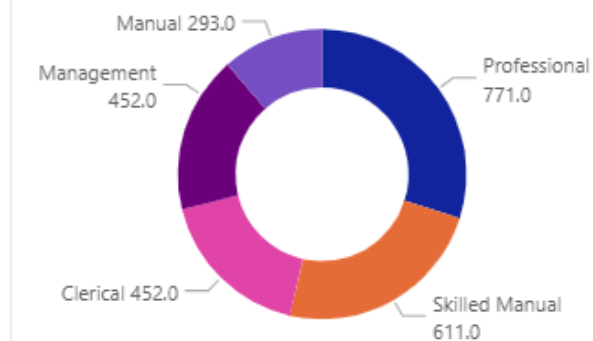
UNIQUE CUSTOMERS

Top Customer By revenue

RAFAEL XU



Total Orders by Customer Income



Top 100 Customers

CustomerKey	Full Name	Orders	Revenue
11090	TREVOR BRYANT	1	3,578.27
11129	JULIA WRIGHT	1	3,578.27
11171	JONATHAN HILL	1	3,578.27
11175	LUIS WANG	1	3,578.27
11191	KRISTI PEREZ	1	3,578.27
11216	JASMINE TORRES	1	3,578.27
11217	NATALIE ADAMS	1	3,578.27
11224	TIFFANY LI	1	3,578.27
11227	MARSHALL CHAVEZ	1	3,578.27
11237	CLARENCE ANAND	1	3,374.96
Total		2630	64,04,933.58

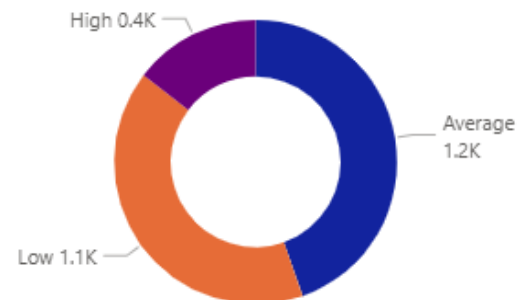
Year

2020

2021

2022

Total Orders by Customer In



352.93

REVENUE PER CUSTOMER

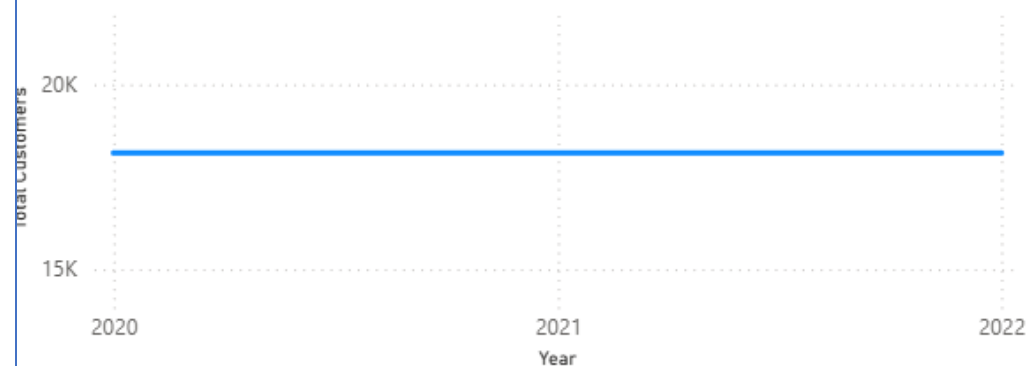
18K

UNIQUE CUSTOMERS

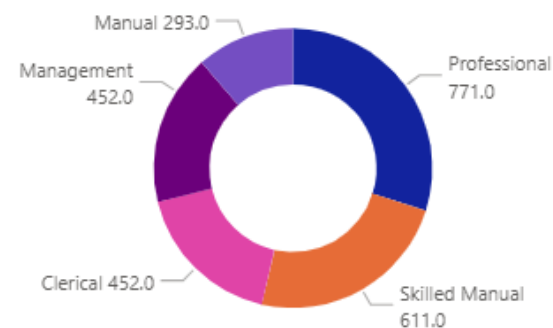
Top Customer By revenue

RAFAEL XU

Total Customers by Year



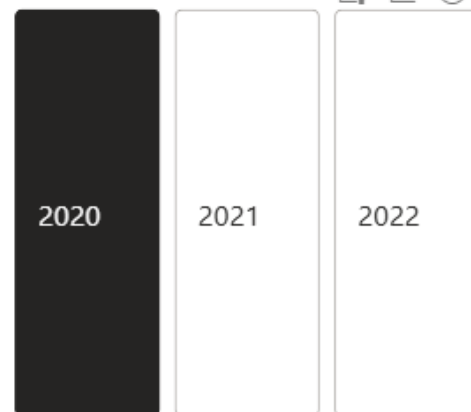
Total Orders by Customer Income



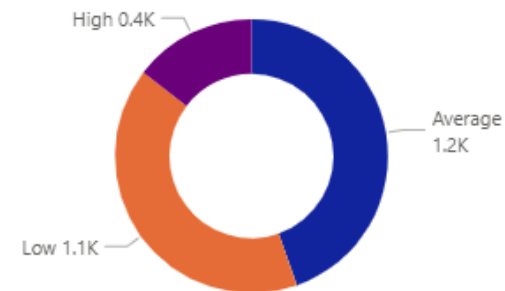
Top 100 Customers

CustomerKey	Full Name	Orders	Revenue
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11216	JASMINE TORRES	1	3,578.27
11217	NATALIE ADAMS	1	3,578.27
11224	TIFFANY LI	1	3,578.27
11227	MARSHALL CHAVEZ	1	3,578.27
11237	CLARENCE ANAND	1	3,374.00
Total		2630	64,04,933.58

Year



Total Orders by Customer Income



352.93

REVENUE PER CUSTOMER

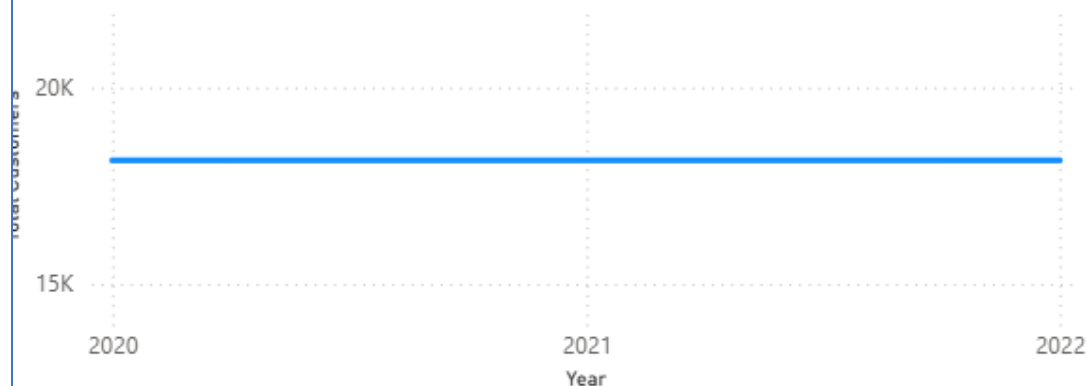
18K

UNIQUE CUSTOMERS

Top Customer By revenue

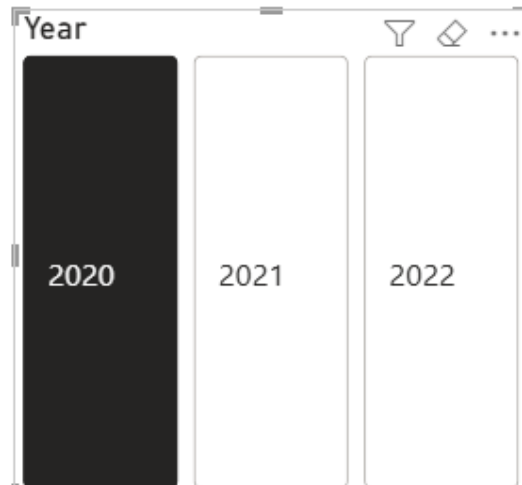
RAFAEL XU

Total Customers by Year

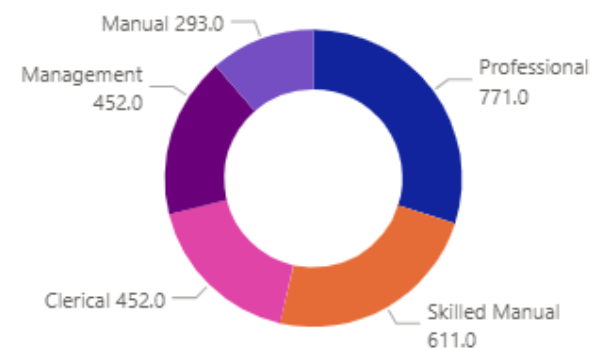


Top 100 Customers

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Total Orders by Customer Income



Total Orders by Customer Income

